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Enhancing the high-temperature performance of high-entropy ceramic/Zr-4 joints via Ni-assisted reactive brazing

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Abstract

Zr-4 alloys exhibit limited oxidation resistance at elevated temperatures, severely restricting their use in extreme environments. High-entropy ceramics (HECs), owing to their exceptional thermal stability and oxidation resistance, have emerged as promising candidates for protective cladding applications. However, reliable joining techniques and a comprehensive understanding of the interfacial phenomena between HECs and Zr-4 alloys are still lacking. In this study, reliable HEC/Zr-4 joints were successfully fabricated via reactive brazing using a pure Ni interlayer. Optimal joining of HEC and Zr-4 was achieved at 1100°C for 10 min, forming a microstructure of HEC/TaC+ZrC/Zr_2Ni+Zr(s, s)/Zr-4. A continuous ZrC layer effectively alleviated thermal expansion mismatch, achieving high shear strengths of 118 MPa at room temperature (RT) and 109 MPa at 800°C. Oxidation tests demonstrated that residual Ni at the interface facilitated the formation of a compact oxide layer, which significantly suppressed oxygen ingress. After oxidation at 900°C for 4h, the joints retained 90% of their initial shear strength, indicating excellent high-temperature stability and oxidation resistance.

Previous Next

Keywords

High-entropy ceramics; Zr-4 alloy; Microstructure evolution; Performance enhancement; Brazing

1. Introduction

As global energy demand continues to escalate, nuclear power is increasingly regarded as a promising alternative to fossil fuels, owing to its high energy density, low carbon emissions, and long-term sustainability [[1], [2], [3]]. Despite these advantages, safety remains a critical concern in nuclear systems, especially the reliability of cladding materials under severe accident conditions [4,5]. Zirconium-based alloys are widely used as fuel cladding in light water reactors, primarily because of their low thermal neutron absorption cross-section [6], strong corrosion resistance [7], and adequate mechanical properties at elevated temperatures [8]. However, the Fukushima Daiichi accident exposed their fundamental vulnerability. During loss-of-coolant accidents (LOCAs), zirconium reacts rapidly and exothermically with high-temperature steam, producing large volumes of hydrogen [[9], [10], [11]]. This greatly increases the risk of core meltdown and radioactive release. Moreover, air ingress after coolant loss further accelerates zirconium oxidation, generating additional heat and seriously compromising the structural integrity of the reactor system [12,13].

To address these challenges, extensive research has focused on the development of accident-tolerant fuel (ATF) cladding systems [[14], [15], [16]], particularly through surface modification of zirconium alloys and the application of advanced ceramic coatings. Compared to surface-modified zirconium alloys, which provide only modest improvements, ceramic coatings offer more robust protection under severe accident conditions [[17], [18], [19]]. However, implementing a continuous ceramic layer over the entire length of a commercial fuel rod presents substantial engineering challenges [20]. In response, research attention is shifting toward more feasible localized reinforcement strategies. These include the fabrication of modular components, such as composite end-plugs with a ceramic cap, and the application of discrete ceramic "armor" patches to high-risk cladding areas [[21], [22], [23], [24]]. Such material-structure-function integration approaches offer a pragmatic pathway to accident tolerance by strategically reinforcing the most vulnerable zones.

However, the performance of conventional binary ceramics under intense radiation remains a concern [[25], [26], [27], [28], [29], [30]]. For instance, helium irradiation of binary carbides like SiC at temperatures ≥700°C readily leads to the formation of plate-like defects that resemble microcracks and increase embrittlement risk [31]. In contrast, recent studies on high-entropy ceramics (HECs) reveal their superior irradiation and oxidation resistance [[32], [33], [34], [35]]. This enhancement is attributed to their pronounced chemical disorder and lattice distortion, which effectively suppress the growth of helium bubbles and irradiation-induced dislocation loops, resulting in reduced irradiation hardening and swelling compared to their binary counterparts [36,37]. For example, Xin et al. [38] reported that (ZrTiNbTaW)C maintained excellent lattice integrity under ion irradiation, with significantly less lattice expansion than ZrC. Zhang et al. [39] further demonstrated the outstanding radiation tolerance of high-entropy carbides through high-temperature helium irradiation. Additionally, the dense microstructure of HECs such as (TiZrNbTaCr)C has been shown to effectively suppress steam penetration and reduce oxidation rates at high temperatures, outperforming binary transition metal carbides [40]. Based on these advantages, HECs are regarded as promising candidates for next-generation nuclear reactor cladding. To fully leverage the complementary strengths of HECs and Zr-4 alloys, establishing a reliable joining method is essential for enhancing the high-temperature oxidation resistance of composite cladding structures.

Nevertheless, joining ceramics to metals remains a significant challenge. This is primarily attributed to the mismatch in thermal expansion coefficients and the inherent brittleness of ceramics, which often leads to interfacial cracking and eventual joint failure. Among the available techniques, brazing is considered a cost-effective and technically feasible method for dissimilar material joining [[41], [42], [43]]. However, conventional Cu- and Ag-based filler metals exhibit insufficient thermal stability, rendering them unsuitable for the demanding service conditions of nuclear reactors [44,45]. Qi et al. [46] brazed SiC to Zr-4 alloy using a Zr-Ti-Cu-Ni amorphous filler, and achieved a joint shear strength of 95 MPa. Fang et al. [47] utilized a Zr-24Ni (at.%) interlayer in the brazing of SiC and Zr-4, which improved the joint strength to 110 MPa. Their findings also showed that Ni-based fillers reduce radiation damage peaks and enhance interfacial compatibility. Similarly, the use of a CoCrFeNiCuSn high-entropy alloy composite interlayer for brazing SiC to Zr has been reported to achieve a high shear strength of 221 MPa at room temperature and 207 MPa at 600°C, further demonstrating the potential of high-entropy systems in joining for nuclear applications [48]. Despite these advances, studies on the joining behavior and high-temperature performance of HECs with Zr-4 alloys remain scarce. Therefore, developing robust HEC/Zr-4 joints with reliable mechanical integrity and oxidation resistance is critical for advancing accident-tolerant nuclear cladding systems.

In this work, reactive brazing of the HEC (TiZrTaNbCr)C and Zr-4 alloy was performed using a pure Ni interlayer to overcome interfacial mismatch and enhance high-temperature reliability. The influence of brazing temperature on interfacial microstructure evolution and joint mechanical properties was systematically characterized at both room and elevated temperatures. Furthermore, the high-temperature oxidation behavior of the joints was evaluated. Based on phase diagram analysis and elemental activity calculations, the formation and evolution of the interfacial eutectic structures during brazing were elucidated. This fundamental study thus offers valuable insights and a scientific basis for the design of not only thermally stable ceramic/metal joints but also for the future realization of the aforementioned localized reinforcement strategies for advanced nuclear fuel cladding systems.

2. Materials and method

2.1. Materials

The HEC (TiZrTaNbCr)C was fabricated via spark plasma sintering (SPS) as reported previously [49], exhibiting high density and excellent high-temperature oxidation resistance. The Zr-4 alloy was supplied by Shenzhen AVIC Special Alloy Co., Ltd., with its chemical composition and physical properties listed in Table 1, Table 2, respectively. The HEC and Zr-4 alloy were machined into block samples with dimensions of $5 \times 5 \times 5\,mm^3$ and $10 \times 10 \times 5\,mm^3$, respectively. Prior to brazing, the joining surfaces of both the HEC and Zr-4 alloy were ground and polished to remove surface oxides and contaminants. Subsequently, all samples were ultrasonically cleaned in acetone for 5 min to eliminate residual particles and grease, then dried for further use.

Table 1. The chemical compositions of Zr-4 alloy (at. %).

Elements	**Sn**	**Fe**	**Cr**	**C**	**Hf**	**Zr**
At. %	1.37	0.20	0.090	0.004	≤0.010	Bal

Table 2. The physical properties of Zr-4 alloy.

Melting point	Density	Elasticity modulus	YS	UTS	CTE
1758°C	6.55g/cm^3	99GPa	350MPa	520MPa	7.2×10^{-6}/K

2.2. Experimental method

A 50μm-thick pure Ni foil was employed as the interlayer to join the (TiZrTaNbCr)C ceramic and the Zr-4 alloy, as illustrated in Fig. 1. The Ni foil was placed between the (TiZrTaNbCr)C ceramic and the Zr-4 alloy to assemble a sandwich structure (Fig. 1(a)). The assembled samples were brazed in a vacuum brazing furnace (VF1600, Zhengzhou Cy Scientific Instrument Co., Ltd, China), with the vacuum pressure maintained at 5×10^{-3}Pa (Fig. 1(b)). The brazing process was carried out at temperatures of 1000°C, 1050°C, 1100°C, and 1150°C, with a holding time of 10min. The heating profile is shown in Fig. 1(d). The shear strength of the brazed joints was tested at RT and 800°C using a high-temperature universal testing machine (AG-Xplus, Shimadzu, Japan) with a loading speed of 0.5mm/min (Fig. 1(c)). The 800°C test was selected to evaluate the mechanical integrity of the joint under a simulated beyond-design-basis accident condition [50]. To evaluate the high-temperature oxidation resistance of the joints, brazed samples were oxidized in a muffle furnace (KSL-1200X-J, Hefei Kejing Materials Technology Co., Ltd, China) at 900°C for 4h, followed by shear strength testing under the same loading conditions.

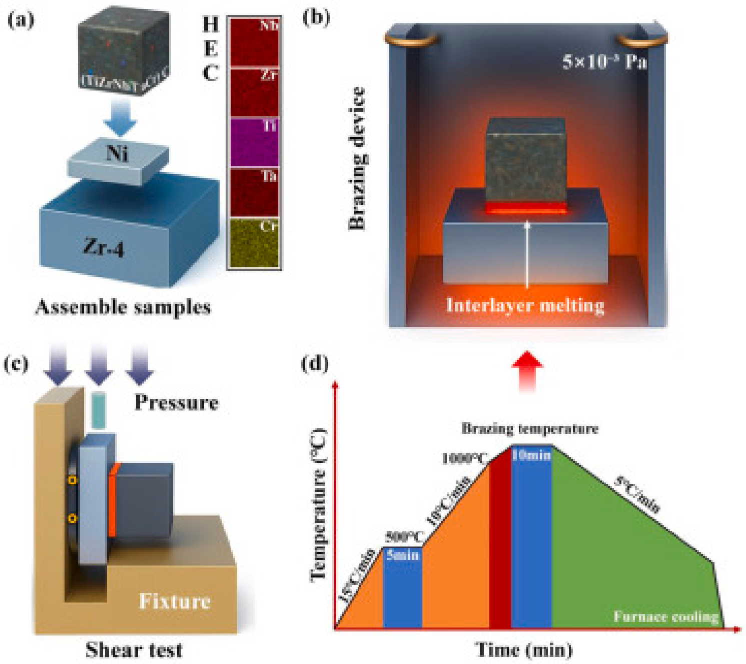

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Fig. 1. Experimental procedure: (a) Schematic of brazing sample assembly, (b) brazing process, (c) schematic diagram of shear strength testing, (d) heating profile.

2.3. Materials characterization

The microstructure and elemental distribution of the (TiZrTaNbCr)C/Zr-4 alloy joints were examined using a field-emission scanning electron microscope (SEM, ZEISS Sigma500, Germany) equipped with energy-dispersive X-ray spectroscopy (EDS). For precise identification of interfacial reaction phases, the joints were ground and polished to expose the bonding interface. Selective etching with aqua regia was applied to remove the Zr-4 alloy matrix and metallic brazing filler, thereby revealing the reaction layer on the ceramic side. Phase identification was performed via X-ray diffraction (XRD, Empyrean, PANalytical, Netherlands). To further investigate the interfacial microstructure and reaction phases, thin-foil specimens were prepared from the reaction layer using focused ion beam (FIB, Helios NanoLab 600i, FEI, USA) milling, followed by characterization with transmission electron microscopy (TEM, Talos F200X, FEI, USA). Additionally, fracture morphologies of the joints after shear testing at RT, elevated temperature, and post-oxidation were observed using SEM.

3. Results

3.1. Typical interfacial microstructure of HEC/Ni/Zr-4 joints

Fig. 2 presents the interfacial microstructure of the joint formed between the HEC ceramic and the Zr-4 alloy, joined using a 50 μm Ni interlayer at 1100 °C. As shown in Fig. 2(a), the brazed seam exhibits a width of approximately 300 μm, exceeding the original Ni thickness. This expansion indicates effective interfacial bonding between Ni and the Zr-4 alloy, accompanied by substantial Ni diffusion into the Zr matrix. To determine the phase composition, EDS point analyses were performed at the locations marked in Fig. 2(b), with the results listed in Table 3. The bright blocky phase near the ceramic interface (Point A) and the dispersed bright particles within the light-gray matrix (Point D) are both enriched in Zr and are identified as Zr solid solution (Zr(s, s)) phases. The dark reaction layer adjacent to the ceramic (Point B) primarily consists of Zr and C, indicating the formation of ZrC. The eutectic region (Point C) features a bright phase with 73.3 at.% Zr, corresponding to Zr_2Ni. Additionally, a small amount of Ni (1.7 at.%), along with the constituent elements of the HEC (Ti, Nb, Ta, Cr), was detected near the ceramic interface (Point E). Meanwhile, the Zr content (34.9 at.%) was significantly higher than the nominal value of the ceramic matrix. This indicates limited elemental interdiffusion occurred during brazing, involving the migration of Ni and Zr (from the metallic side) towards the ceramic side.

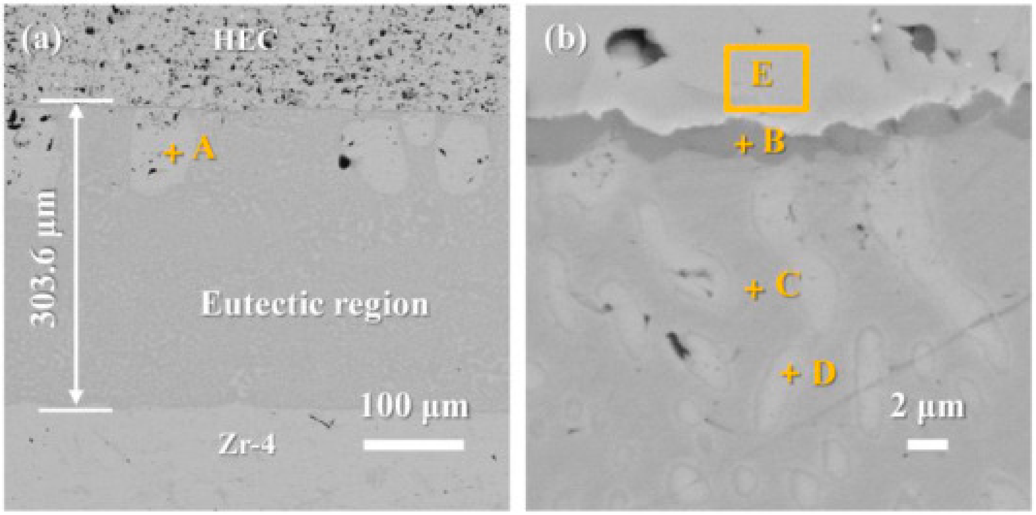

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Fig. 2. Interfacial microstructure of HEC ceramic/Zr alloy joint with Ni interlayer: (a) overall morphology of the joint, (b) enlarged view of the ceramic-side region.

Table 3. The element composition of the marked zone in Fig. 2.

Point	Ni	Cr	Ti	Zr	Nb	Ta	C
A	2.2	–	–	97.8	–	–	–
B	1.4	–	–	88.2	–	–	10.4
C	26.7	–	–	73.3	–	–	–
D	5.3	–	–	94.7	–	–	–
E	1.7	3.9	14.1	34.9	22.9	19.4	3.2

Fig. 3 shows the elemental distribution across the joint interface brazed at 1100°C. The results indicate that Ni diffused extensively into the Zr-4 alloy and was detected not only in the eutectic region but also in the surrounding areas, demonstrating its high diffusivity at elevated temperatures. In contrast, ceramic-side elements (Cr, Ta, Ti) exhibited negligible diffusion toward the metallic side and remained concentrated within the HEC matrix.

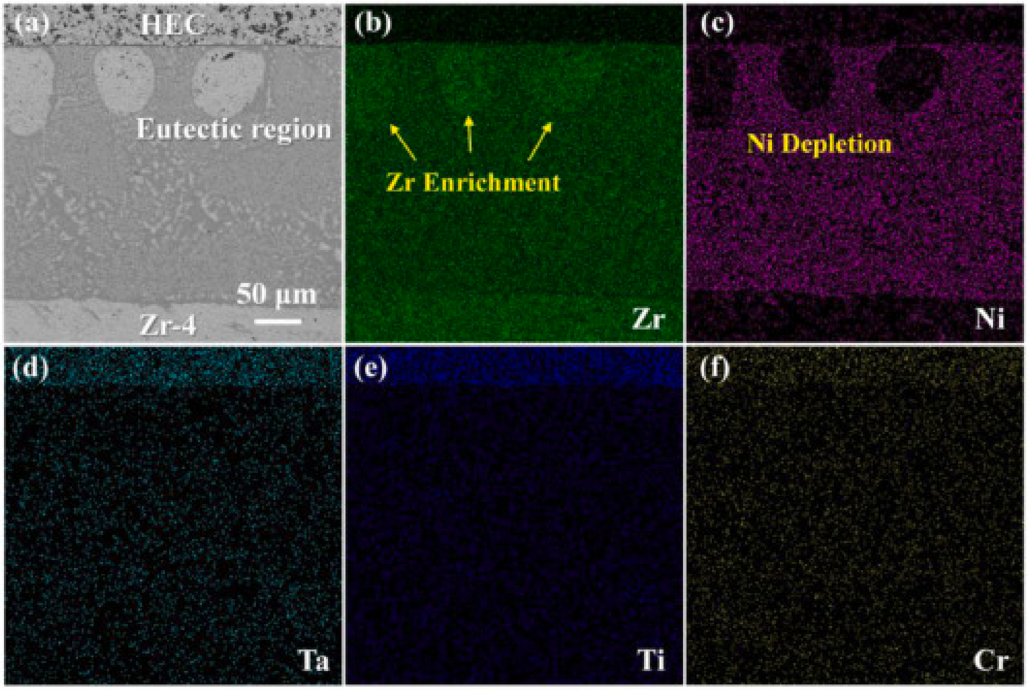

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Fig. 3. Elemental distribution at the interface of HEC/Ni/Zr-4 joint brazed at 1100°C: (a) microstructural morphology of the joint, (b) Zr, (c) Ni, (d) Ta, (e) Ti, (f) Cr.

The phase composition of the joint interface was further confirmed by XRD analysis, as shown in Fig. 4. The sampling location is illustrated in Fig. 4(a). As presented in Fig. 4(b), the reaction layer on the ceramic surface primarily consists of ZrC, while the brazed seam mainly contains Zr_2Ni and Zr phases. Fig. 4(c) reveals a slight shift in the diffraction peaks of the HEC, suggesting that some Ni atoms diffused into the ceramic and partially substituted metal lattice positions during the brazing process [51,52].

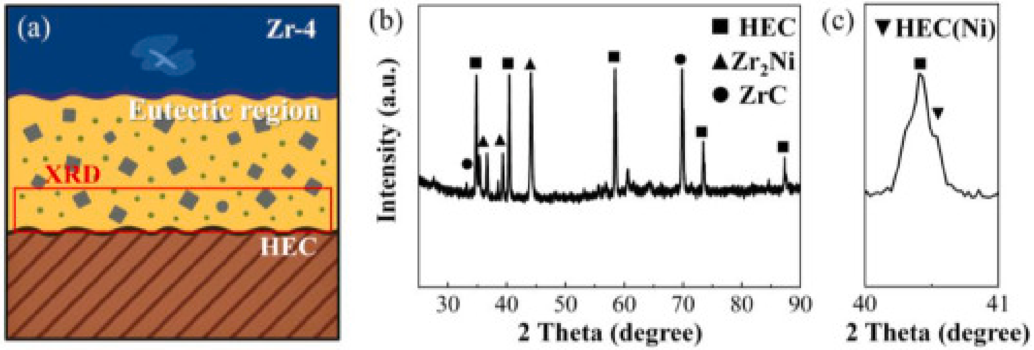

Fig. 4. XRD patterns of the reaction layer in HEC ceramic/Zr alloy joint with Ni interlayer: (a) sampling location, (b) 2θ=25–90°, (c) 2θ=40–41°.

Fig. 5 presents the TEM interfacial microstructure and selected area electron diffraction (SAED) patterns of the reaction layer in the HEC/Zr-4 alloy joint with a Ni interlayer. As shown in Fig. 5(a), the reaction layer exhibits a continuous multiphase structure, with the interface consisting sequentially of HEC, TaC, ZrC, Zr_2Ni, and Zr(s, s), as delineated by the dashed lines. The formation of a dense and continuous reaction layer between the HEC and Zr-4 alloy indicates good interfacial bonding Fig. 5(b–f) display the SAED patterns corresponding to the crystal orientations of HEC ([0-11]), TaC ([−101]), ZrC ([[1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21]]), Zr_2Ni ([111]), and Zr(s, s) ([010]), respectively [[53], [54], [55]]. Based on the microstructural observations and elemental distributions, the TaC phase is attributed to local Ta enrichment during brazing. The ZrC phase forms the main reaction layer on the ceramic side. The Zr_2Ni phase originates from the reaction between Ni and Zr, while the Zr(s, s) phase is identified as a Zr-based solid solution.

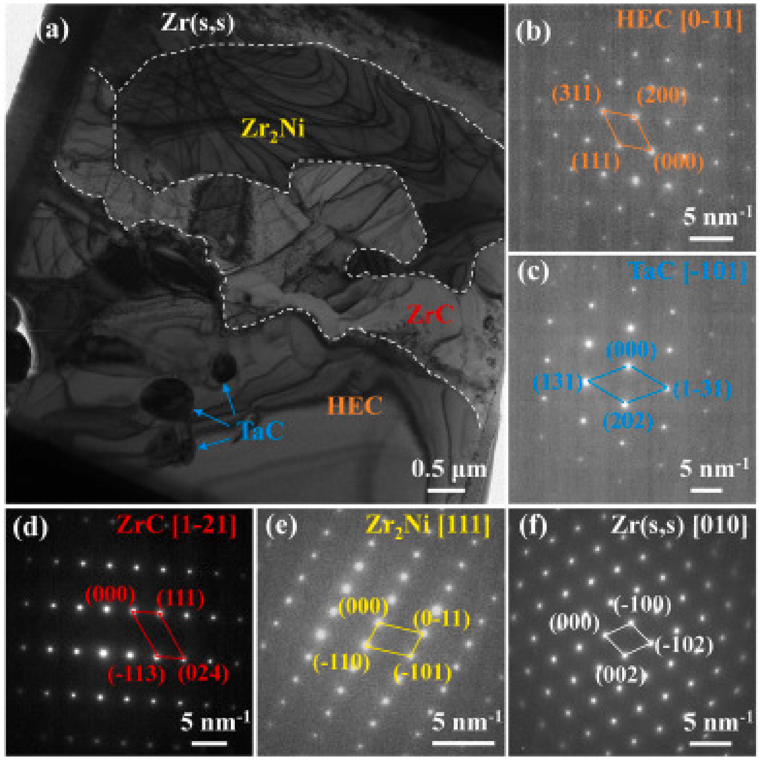

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Fig. 5. TEM analysis of the reaction layer in the HEC/Ni/Zr-4 joint: (a) bright-field image of the interfacial structure, (b) HEC [0-11], (c) TaC [−101], (d) ZrC [[1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21]], (e) Zr_2Ni [111], (f) Zr (s,s) [010].

Fig. 6 further illustrates the interfacial details between the HEC and the ZrC layer. The bright-field TEM image in Fig. 6(a) shows a continuous ZrC layer forms at the interface, establishing a well-bonded interface with the HEC. As depicted in Fig. 6(b), the (111) planes of HEC and ZrC are aligned in parallel with an interplanar angle of approximately 8°. The measured lattice spacings of HEC and ZrC are 0.256 nm and 0.260 nm, respectively, indicating a low lattice mismatch that promotes strong interfacial bonding [56]. The elemental distribution in the interfacial region is shown in Fig. 6(c), where significant Zr enrichment was observed at the interface, and C was uniformly distributed. This indicates that Zr from the metallic side reacted with C from the ceramic side to form ZrC, with limited involvement of Ni in the interfacial reaction. Additionally, the principal elements of the HEC (Nb, Ti, Cr, and Ta) remain concentrated within the ceramic region without appreciable diffusion, indicating compositional stability of the ceramic matrix near the joint.

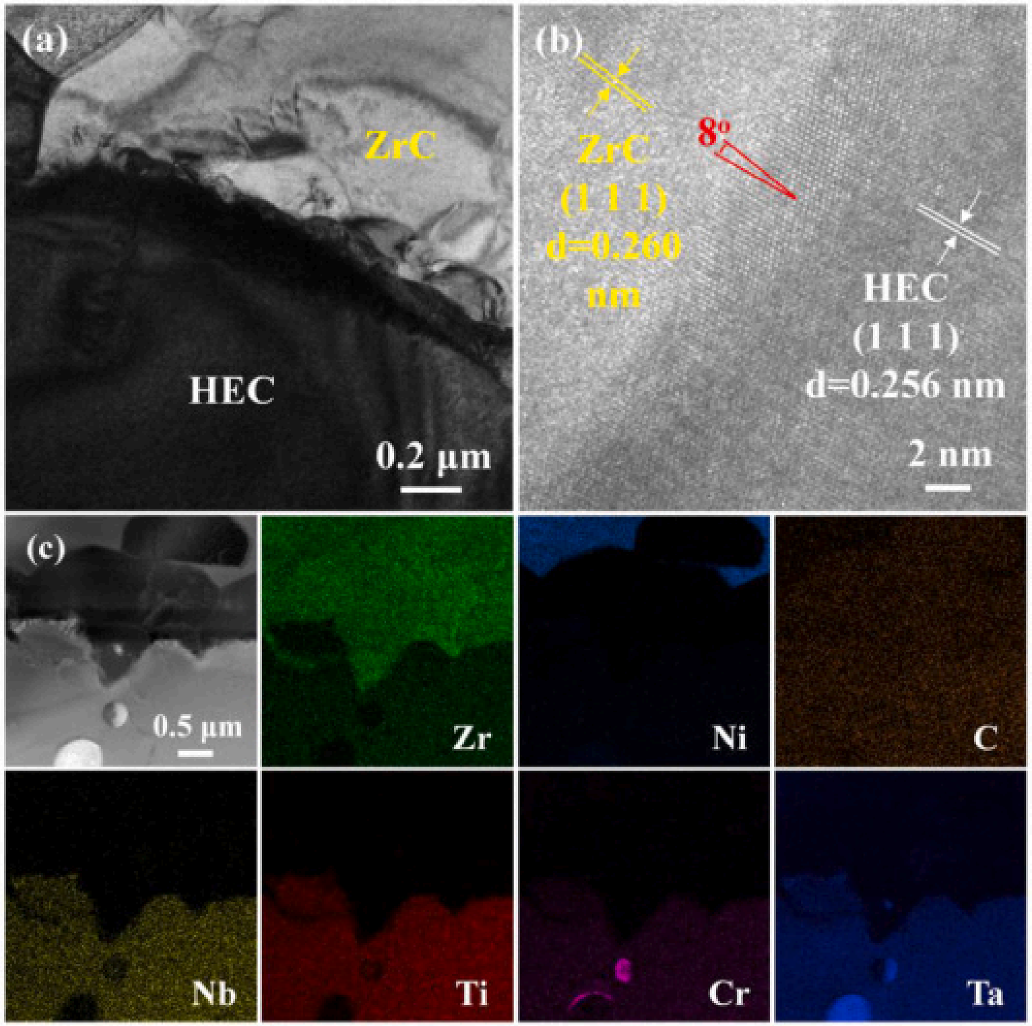

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Fig. 6. TEM images of the interfacial microstructure between HEC and ZrC: (a) bright-field image of the interface, (b) high-resolution image, (c) elemental distribution at the interface.

3.2. Effect of brazing temperature on the microstructural evolution of HEC/Ni/Zr-4 joints

Fig. 7 presents the surface microstructures of joints brazed at 1000°C, 1050°C, 1100°C, and 1150°C. As shown in Fig. 7(a–d), the width of the diffusion zone increases significantly with temperature, from 36.9μmat 1000°C to 282.3μmat 1100°C. However, when the temperature is further raised to 1150°C, the diffusion zone narrows considerably. Under this temperature, only a limited amount of Zr_2Ni remains at the interface between the ZrC layer and the Zr-4 substrate, while the joint region is primarily composed of the Zr(s, s) phase. Fig. 7(e–f) further indicate that the thickness of the ZrC reaction layer increases with temperature up to 1100°C and then remains nearly constant.

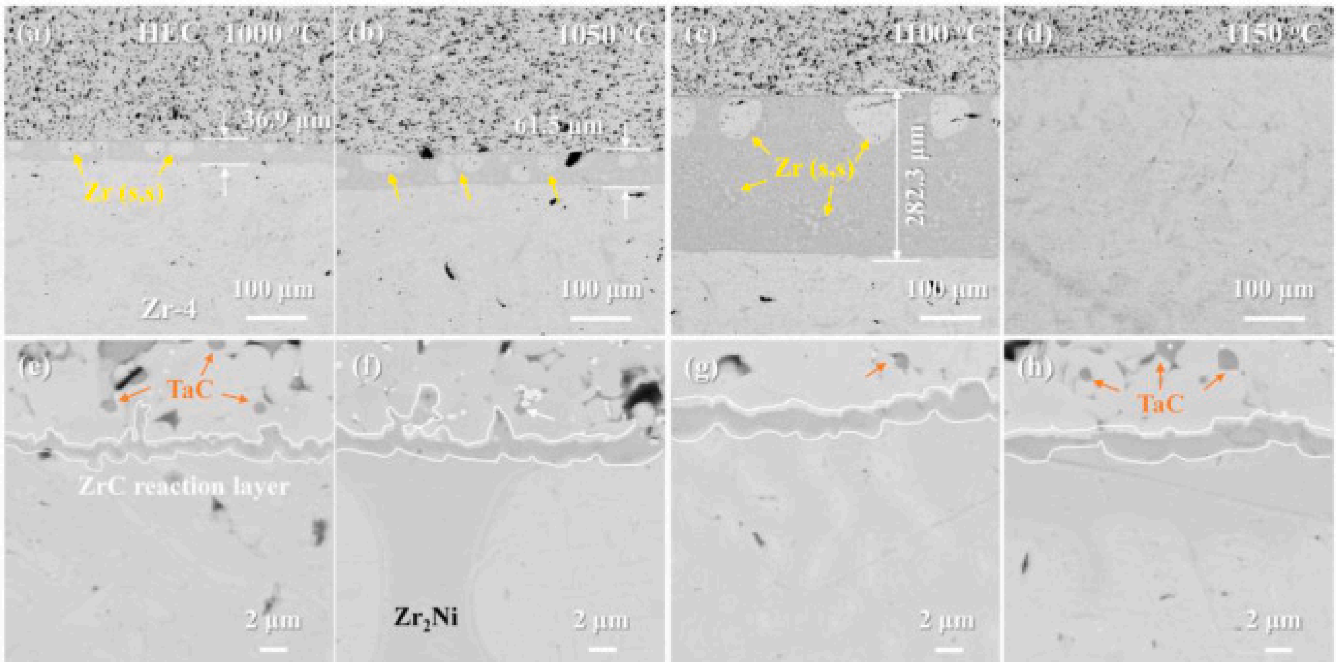

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Fig. 7. Effect of brazing temperature on the microstructure of HEC/Ni/Zr-4 joints: (a, e) 1000°C, (b, f) 1050°C, (c, g) 1100°C, (d, h) 1150°C.

The elemental distribution in the HEC/Ni/Zr-4 joint brazed at 1150°C was further investigated, as shown in Fig. 8. Fig. 8(a) presents the cross-sectional morphology of the joint, while Fig. 8(b–f) display the corresponding elemental maps of Ni, Zr, Ta, Cr, and Ti, respectively. The results reveal that Ni is dispersed within the Zr alloy region with slight local enrichment, and Zr is uniformly distributed without the formation of distinct eutectic structures. In contrast, Ta, Cr, and Ti remain concentrated in the ceramic region, showing no appreciable diffusion toward the metallic side. Overall, the interfacial structure of the joint brazed at 1150 °C consists of an HEC/ZrC/Zr configuration, in which a continuous and compact ZrC reaction layer is the sole interfacial product.

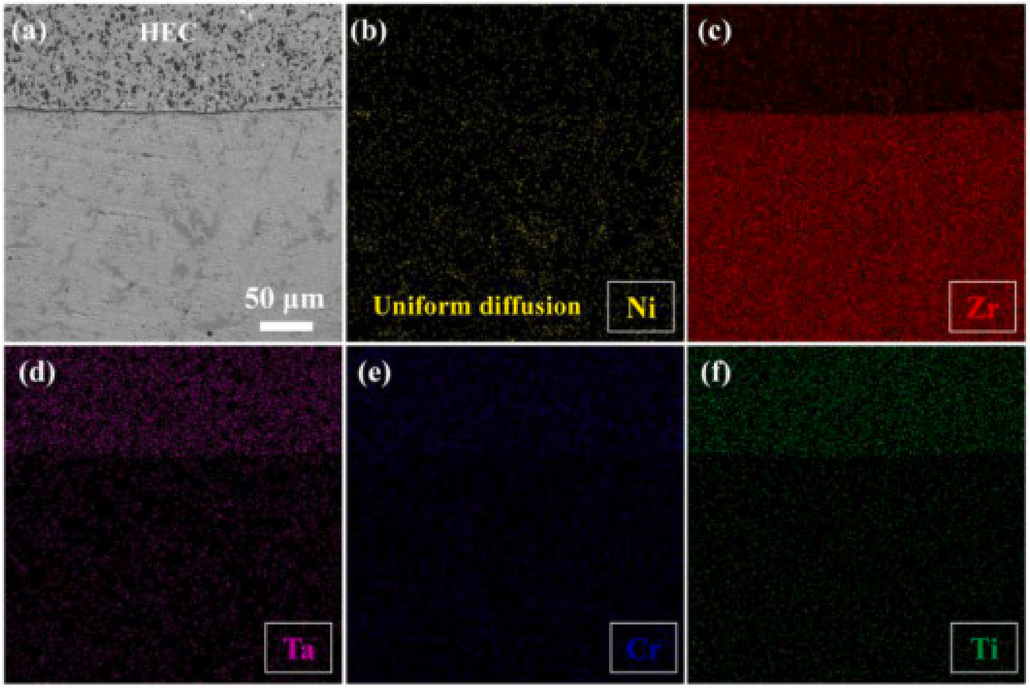

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Fig. 8. Elemental distribution at the interface of the HEC/Ni/Zr-4 joint brazed at 1150 °C: (a) cross-sectional morphology of the joint, (b) Ni, (c) Zr, (d) Ta, (e) Cr, (f) Ti.

3.3. Effect of brazing temperature on the mechanical properties of HEC/Ni/Zr-4 joints

Fig. 9 illustrates the variation in shear strength at RT and 800 °C. Shear strength initially increased with brazing temperature, reaching a maximum at 1100 °C (118 MPa at RT and 109 MPa at 800 °C), followed by a decline at 1150 °C. Notably, at all brazing conditions, the high-temperature shear strength remained only slightly lower than the corresponding room-temperature value, indicating good mechanical stability and thermal reliability of the joints under elevated service conditions.

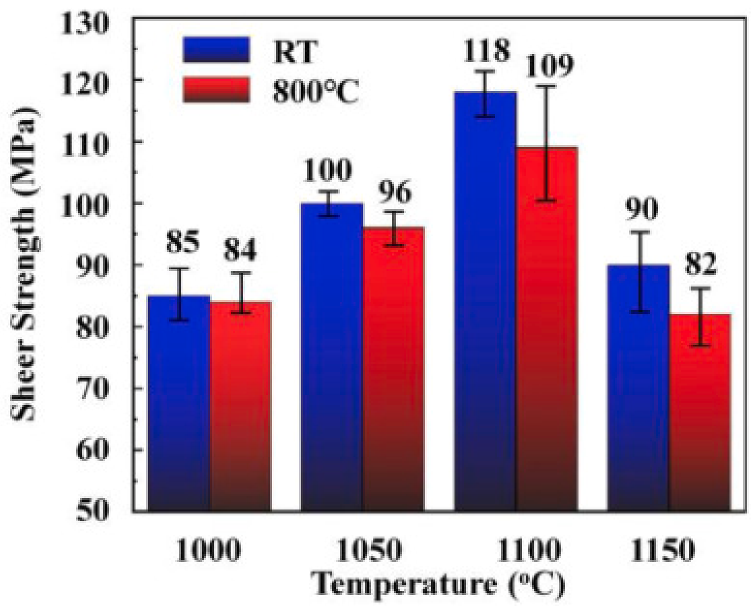

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Fig. 9. Shear strength of HEC/Ni/Zr-4 joints at RT and 800°C under different brazing temperatures.

Fig. 10 presents the fracture morphologies of the joints after RT shear testing. For specimens brazed at 1000°C, 1050°C, and 1150°C, fracture occurred along the interfacial reaction layer and propagated toward the ceramic side, resulting in relatively flat fracture surfaces with features characteristic of brittle failure (Fig. 10(a), (b), and 10(d)). In contrast, when the brazing temperature was increased to 1100°C, the fracture path shifted further into the ceramic region (Fig. 10(c)). The corresponding fracture surface exhibited fine granular features (Fig. 10(c_1)), suggesting a notable enhancement in interfacial bonding strength. The fact that the fracture path propagated into the ceramic substrate at 1100°C (Fig. 10(c)) provides direct evidence for the superior interfacial bonding achieved under this optimal parameter, which is consistent with the peak shear strength shown in Fig. 9. This shift in the primary failure site indicates that the robust metallurgical bond, facilitated by the continuous ZrC layer and the ductile eutectic structure, became stronger than the HEC ceramic itself. Consequently, the joint's load-bearing capacity was ultimately determined by the intrinsic strength of the ceramic, leading to the maximum shear strength.

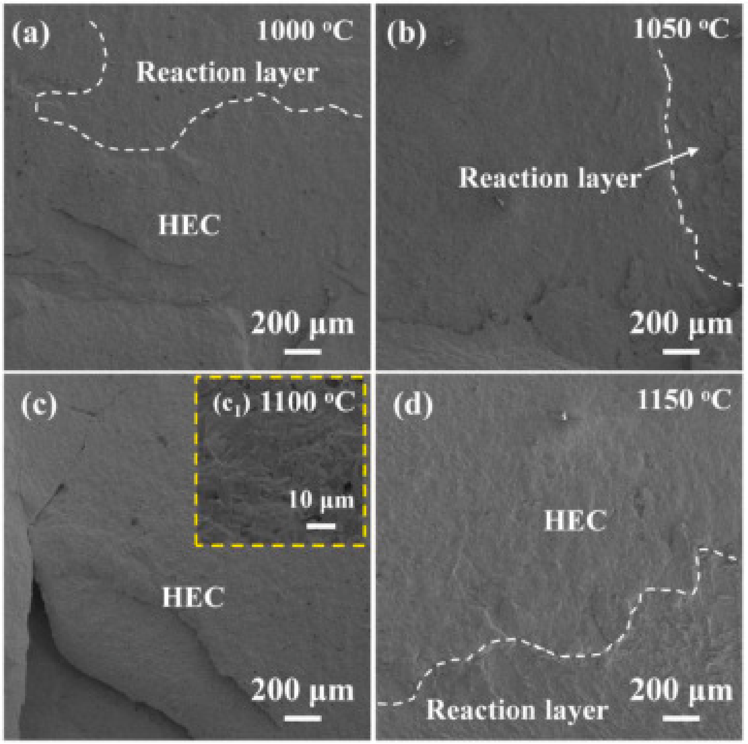

Fig. 10. Fracture morphology of HEC/Ni/Zr-4 joints at RT under different brazing temperatures: (a) 1000°C, (b) 1050°C, (c) 1100°C, (d) 1150°C.

Fig. 11 shows the fracture morphology of the joint brazed at 1100°C after shear testing at 800°C. The fracture predominantly occurred near the interface between the ceramic side and the Zr-Ni eutectic region. On the metallic side, pronounced tear ridges were observed on the fracture surface (Fig. 11(b)), accompanied by localized stepped shear features. These features indicate that the joint underwent a certain degree of plastic deformation at elevated temperature. The overall fracture mode was identified as a mixed ductile-brittle failure.

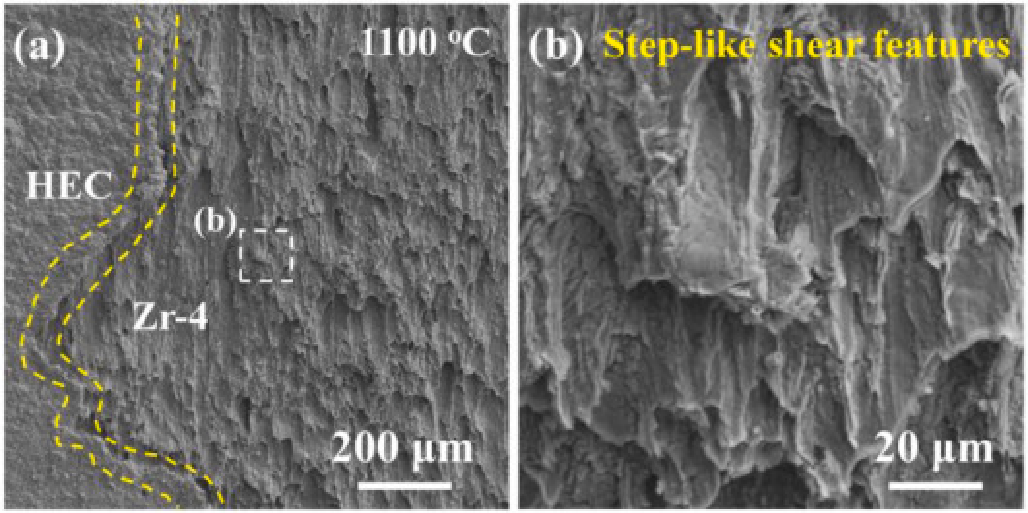

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Fig. 11. High-temperature fracture morphology of HEC/Ni/Zr-4 joints: (a) low-magnification view of the metal-side fracture, (b) high-magnification view of the metal-side fracture.

4. Discussion

4.1. Temperature-dependent diffusion and interfacial evolution in Ni-assisted brazed HEC/Zr-4 joints

The Zr-Ni phase diagram and the thermodynamic behavior of Zr activity are key factors governing the interfacial evolution in HEC/Ni/Zr-4 joints. These aspects are summarized in Fig. 12, which illustrates both the phase equilibrium and Zr activity trends [57]. As shown in Fig. 12(a), a eutectic reaction occurs at a Zr content of approximately 76at.%, with a eutectic temperature of 975 °C, notably lower than the brazing temperatures employed in this study (1000–1150 °C). This low eutectic point promotes sufficient liquid-phase formation and favorable filler fluidity, essential for achieving high joint density and good interfacial wetting [47].

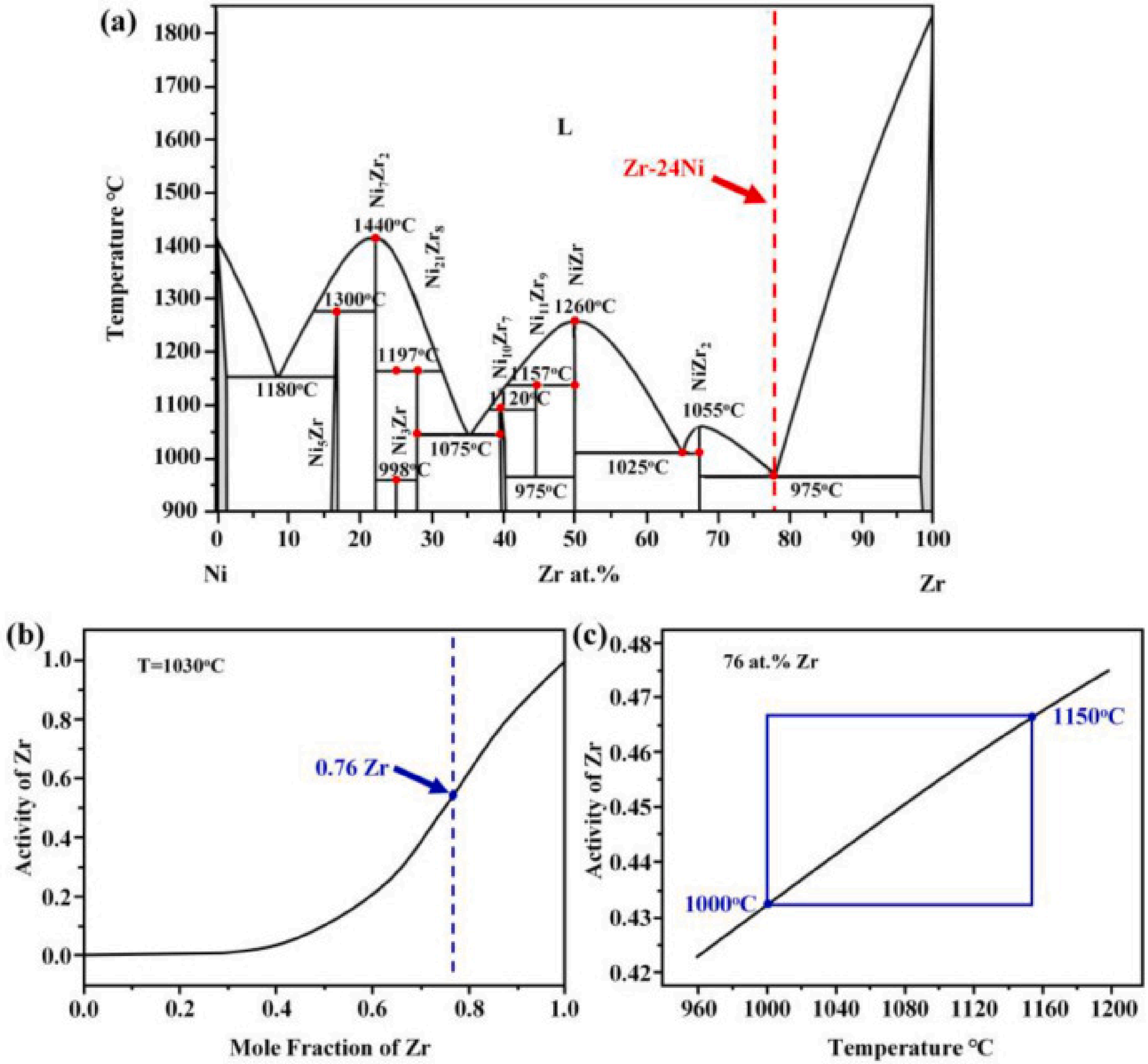

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Fig. 12. Zr-Ni phase diagram and thermodynamic behavior of Zr activity: (a) Zr-Ni binary phase diagram, (b) relationship between Zr activity and Zr content at 1030°C, (c) variation of Zr activity with temperature in the Zr-Ni system.

The diffusion behavior and interfacial reactions are significantly influenced by the activity of Zr. As shown in Fig. 12(b–c), the calculated Zr activity increases steadily with rising Zr content at 1030°C, and exhibits a sharp increase near the eutectic composition [58]. Moreover, within 1000–1150°C, Zr activity increases with temperature, providing a strong driving force for Zr diffusion toward the ceramic interface and promoting a continuous, dense ZrC reaction layer [59].

The interfacial evolution at different brazing temperatures is illustrated in Fig. 13. At 1000–1050°C (Fig. 13(a and b)), the diffusion of Ni was relatively limited, and the migration of Zr toward the brazing seam and ceramic interface was insufficient. As a result, the extent of interfacial reactions was restricted. The joints predominantly consisted of locally enriched Zr_2Ni eutectic phases, a narrow Zr(s, s) region, and a thin ZrC reaction layer, leading to poor interfacial continuity and limited mechanical performance [60].

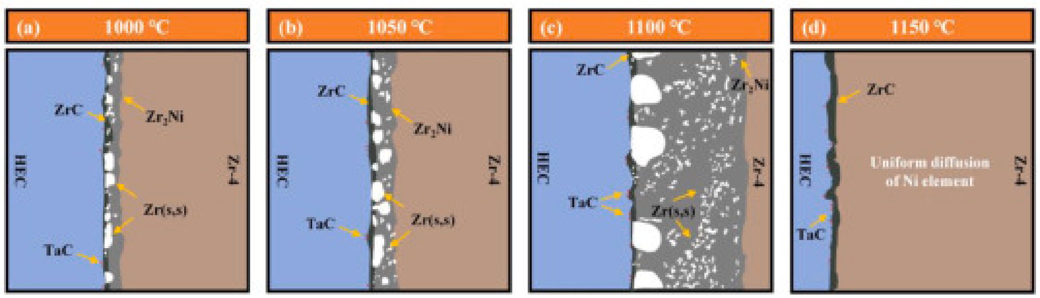

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Fig. 13. Schematic diagrams of joint microstructures at different brazing temperatures: (a) 1000°C, (b) 1050°C, (c) 1100°C, (d) 1150°C.

When the brazing temperature increased to 1100°C (Fig. 13(c)), the diffusion rates of Ni and Zr were significantly enhanced. Sufficient Zr diffusion toward the ceramic side facilitated the formation of a continuous ZrC reaction layer. Meanwhile, a broad Zr_2Ni eutectic region and a Zr(s, s) transition zone formed, effectively buffering residual stress and mitigating thermal expansion mismatch [61]. The brazing seam exhibited good densification and compositional uniformity, and shear strength testing identified 1100°C as the optimal brazing temperature.

At 1150°C (Fig. 13(d)), further elevation in temperature led to widespread dispersion of Ni throughout the joint and a substantial reduction in the Zr_2Ni phase content, with the composition approaching homogenization. The improved fluidity of the liquid phase promoted the expulsion of the eutectic melt, resulting in a simplified interfacial configuration of HEC/ZrC/Zr. However, the disappearance of the metallic buffer zone increased stress localization, ultimately reducing the mechanical integrity of the joint [62].

Collectively, the brazing temperature significantly affects the diffusion kinetics and the evolution of the interfacial microstructure. The optimal brazing temperature of 1100°C offers favorable liquid-phase fluidity, sufficient elemental diffusion, and effective interfacial stress buffering. The thermodynamic characteristics of the Zr-Ni system, particularly the variation in Zr activity with composition and temperature, play a central role in promoting interfacial bonding and ensuring the reliability of the resulting joints.

4.2. Bidirectional diffusion and multiphase reactions governing HEC/Ni/Zr-4 interface formation

The elemental diffusion behavior and interfacial reaction mechanism of the HEC/Ni/Zr-4 joint brazed at 1100°C are illustrated in Fig. 14. Pronounced bidirectional diffusion and multiphase reactions governed interfacial evolution. Initially, Zr originating from the Zr-4 alloy, due to its high chemical reactivity, preferentially diffused toward both the Ni interlayer and the ceramic region [63]. Meanwhile, Ni diffused into both the ceramic and the Zr alloy, although its penetration depth was relatively limited. The high-entropy elements in the HEC ceramic, such as Ti, Nb, Cr, and Ta, exhibited low diffusivity and were largely retained within the ceramic matrix.

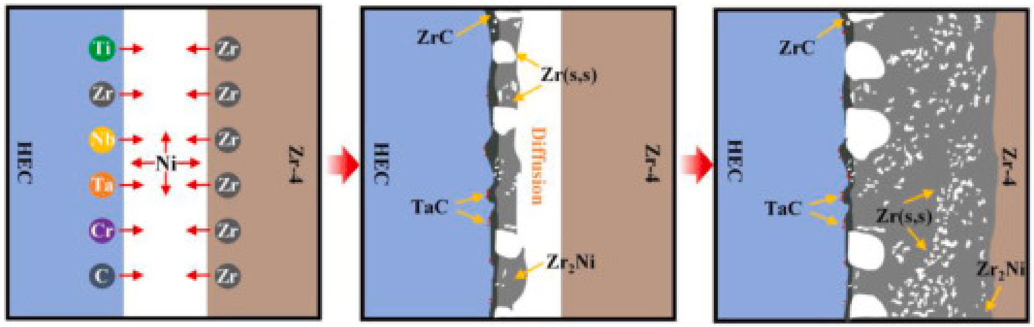

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Fig. 14. Schematic diagram of elemental diffusion and interfacial reaction mechanism at 1100°C.

As diffusion progressed, Zr preferentially reacted with carbon at the ceramic interface, forming a continuous and dense ZrC reaction layer that gradually enhanced interfacial bonding. Locally, Ta became enriched near the interface and reacted with carbon to generate a small amount of TaC. The distinct microstructural manifestations of ZrC and TaC are governed by their respective thermodynamic and kinetic pathways. Gibbs free energy calculations (Fig. 15) reveal a substantially stronger driving force for ZrC formation compared to TaC, which, in conjunction with the abundant Zr supply, promotes the development of a continuous ZrC layer. In contrast, the discrete TaC particles originate from sites of local Ta enrichment, as directly evidenced by EDS and TEM analyses (Fig. 3, Fig. 5, Fig. 6). Within these localized regions, the elevated chemical activities of Ta and C provide the necessary thermodynamic impetus for TaC nucleation, enabling its precipitation alongside the dominant ZrC phase. This multiphase interfacial structure collectively contributes to the joint's mechanical integrity and thermal stability [64]. Within the brazing seam, Ni and Zr underwent eutectic reactions, leading to the formation of the Zr_2Ni intermetallic compound and a Zr-based solid solution [47]. The resulting liquid phase exhibited favorable flowability, enabling complete wetting of the ceramic surface and promoting uniform elemental distribution and interfacial densification.

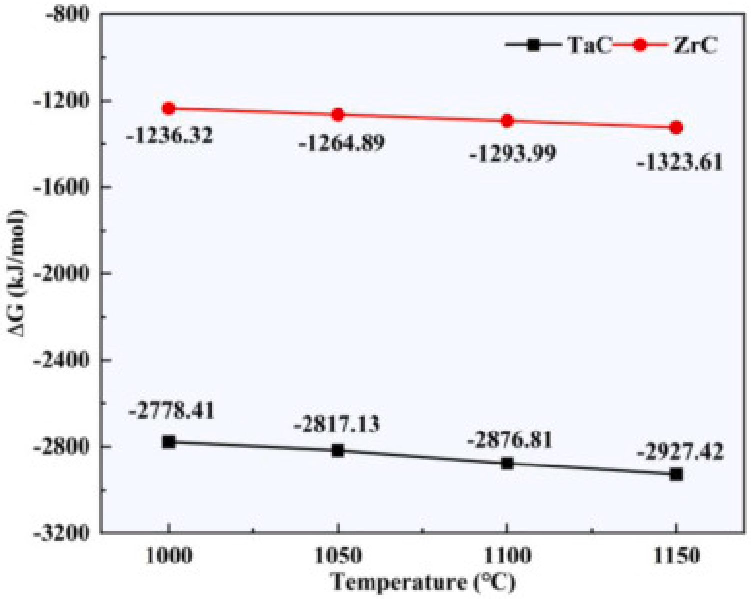

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Fig. 15. Calculation of Gibbs free energy of main products of brazed joints.

In the later stage of diffusion, the migration of both Zr and Ni further intensified. The Zr_2Ni eutectic region continued to expand, while the metallic transition layer effectively buffered the thermal expansion mismatch between the ceramic and the metal [65]. Meanwhile, the ZrC reaction layer further thickened, and the interfacial structure gradually stabilized. At this stage, interfacial evolution was predominantly governed by elemental diffusion, and the synergistic effect of the metallic buffer zone and the reaction layer significantly improved the joint bonding strength.

Overall, the interfacial reaction process was primarily driven by the high-activity diffusion of Zr. The moderate diffusion and eutectic reactions of Ni ensured wetting and compositional transitions. Local enrichment of Ta led to the formation of a limited amount of TaC, which further contributed to the mechanical integrity and thermal stability of the joint. In contrast, Ti, Nb, and Cr exhibited low diffusivity and did not participate in the formation of interfacial phases. The resulting reaction pathway was well-regulated, enabling the formation of a robust metallurgical bond between the HEC and the Zr-4 alloy.

4.3. Oxidation resistance and structural stability of Ni-assisted brazed HEC/Zr-4 joints

To evaluate the high-temperature oxidation behavior and structural integrity of the HEC/Ni/Zr-4 joints, isothermal oxidation tests were conducted at 900°C for 4h on samples brazed at 1100°C [66]. The cross-sectional microstructure after oxidation is shown in Fig. 16. The results indicate that the internal structure of the joint remained stable, with no significant changes observed in the brazing seam. The interfacial structure and the high-temperature diffusion layer effectively suppressed oxygen penetration. A dense Ni-O oxide layer with a thickness of approximately 20μm was observed at the brazing fillet due to the local oxidation of residual Ni [67]. In contrast, a thicker oxidation layer of approximately 150μm was observed on the HEC ceramic surface. The Zr-4 substrate exhibited the most severe oxidation, characterized by a porous and cracked oxide scale exceeding 300μm in thickness and showing a clear tendency toward spallation. This indicates that the Zr-4 alloy suffered the most severe oxidation among the three regions.

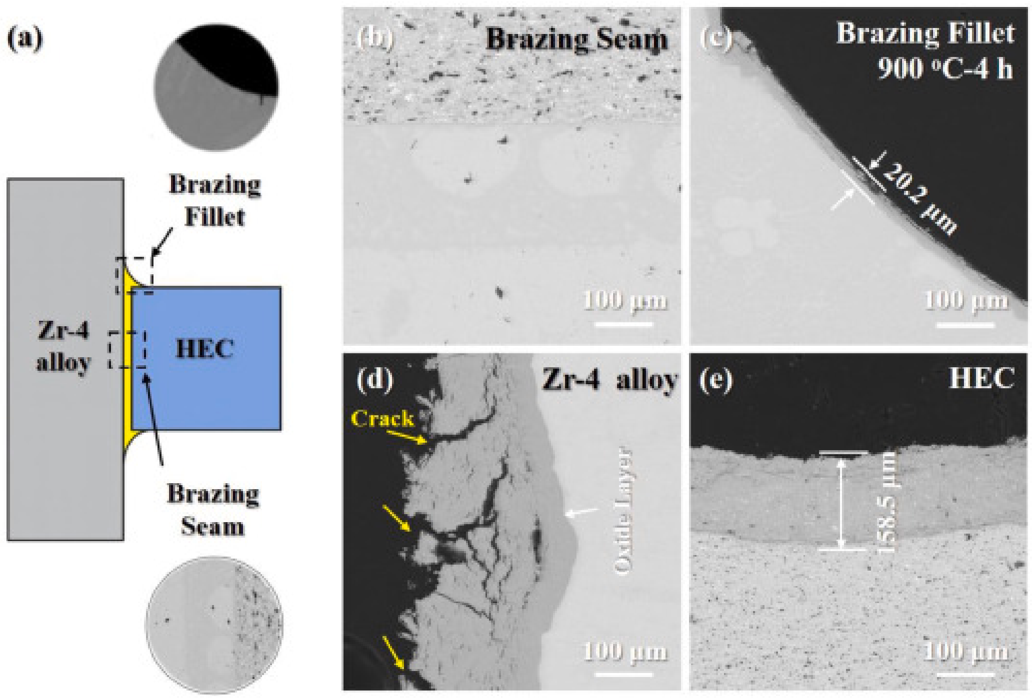

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Fig. 16. Cross-sectional microstructure of the HEC/Ni/Zr-4 joint after oxidation: (a) schematic diagram of the joint cross-section, (b) brazing seam region, (c) brazing fillet, (d) oxidized cross-section of the Zr-4 alloy, (e) surface of the HEC ceramic.

Comparative analysis demonstrates that synergistic protection by the Ni-O film and HEC ceramic limited oxygen ingress into the brazing seam, suppressing overall oxidation. Post-oxidation fracture morphology (Fig. 17) resembled the unoxidized joint, exhibiting brittle failure initiated at the reaction layer. Shear strength retained 106 MPa (approximately 90% of the original value), with strength reduction primarily attributed to ceramic surface degradation reducing the effective bonding area [48]. The brazing seam and metal side maintained structural integrity and load-bearing capacity.

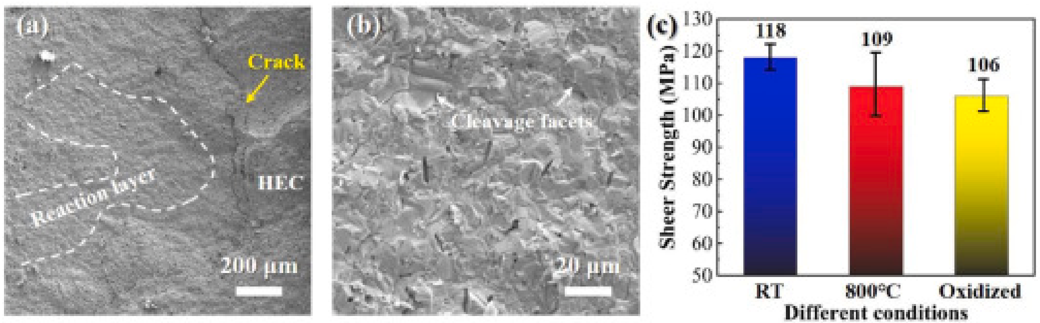

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Fig. 17. Performance of the joint after high-temperature oxidation: (a) low-magnification fracture morphology on the metal side, (b) high-magnification microstructure at the reaction layer, (c) shear strength after high-temperature oxidation.

In conclusion, Ni-interlayered HEC/Zr-4 joints exhibit exceptional oxidation resistance and mechanical stability at high temperatures. Synergistic effects of the continuous ZrC layer, optimized interfacial structure, and dense Ni-O film collectively suppressed oxygen penetration and preserved joint strength.

5. Conclusion

In this study, heterogeneous joining between HEC (TiZrTaNbCr)C and Zr-4 alloy was successfully achieved via Ni interlayer reactive brazing. The interfacial microstructure, mechanical performance, and high-temperature oxidation resistance of the joints were systematically investigated. The main conclusions are as follows:

(1) A dense joint with effective interfacial bonding was achieved using a 50 μm pure Ni interlayer, which fully diffused during brazing and facilitated the formation of a continuous metallic transition zone.

(2) The optimal brazing temperature was identified as 1100 °C, where the joint exhibited a multiphase interfacial structure (HEC/TaC/ZrC/Zr_2Ni/Zr(s, s)) and demonstrated excellent shear strengths of 118 MPa at RT and 109 MPa at 800 °C.

(3) The formation of a continuous ZrC reaction layer and a Zr_2Ni eutectic region effectively alleviated thermal expansion mismatch, thereby enhancing interfacial bonding strength and structural stability.

(4) After oxidation at 900 °C for 4 h, the joints retained approximately 90% of the original shear strength (106 MPa), confirming exceptional oxidation resistance and mechanical reliability.

CRediT authorship contribution statement

Jinze Chi: Writing – original draft. **Pengguan Wang:** Funding acquisition. **Pengcheng Wang:** Writing – review & editing. **Zhaoyi Pan:** Conceptualization. **Yaotian Yan:** Resources. **Xiaoguo Song:** Data curation. **Haiyan Chen:** Visualization. **Wenya Li:** Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Recommended articles

Data availability

Data will be made available on request.

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